

EDUCATION **Stanford University** *Visiting Researcher*

& TRAINING Advisors: Thang Luong (Google Deepmind) & Jeff Glenn (Stanford Medicine)

University of Massachusetts Amherst *B.S. '24 in Computer Science*

Advisors: Bruno Castro da Silva (Computer Science)

RESEARCH My research focuses on generative modeling and decision-making, particularly diffusion models and reinforcement learning, with applications in computational biology. It supports discovery by integrating generative and decision-making methods. Below are selected papers:

- NeurIPS'26* 1. **Dang** et al. Plausible Biomolecular Structure Prediction via Physics-informed Reinforcement Learning.
 • Achieved SOTA in structure prediction fidelity and affinity via RL with physics-based rewards.
 • *Paper*: arXiv, *Blog*: π -Fold, *Code*: nti/pifold
- ICML WS'25* 2. **Dang** et al. Preferential Multi-Objective Bayesian Optimization for Drug Discovery.
 • Recovered 16/37 EGFR and 37/58 DRD2 drugs from 100K ligands via chemist-guided screening.
 • *Paper*: arXiv, *Code*: tai-dang11/cheapvs
- EACL'23* 3. **Dang** et al. Enriching Biomedical Knowledge for Low-resource Language Through Translation.
 • SOTA in Vietnamese biomedical benchmark and high-quality Vietnamese MedNLI dataset.
 • *Paper*: eacl, *Code*: vietai/ViPubmed
- Preprint'22* 4. MTet: Multi-domain Translation for English and Vietnamese.
 • *Details*: SOTA in English-Vietnamese translation and high-quality multi-domain bilingual corpus.
 • *Paper*: arXiv, *Code*: vietai/mTet
- COLING'25* 5. AURORA-M: Open Source Continual Pre-training for Multilingual Language and Code.
 • Developed a 15B open-source multilingual model continually pre-trained on code and text.
 • *Paper*: coling, *Model*: huggingface.co/aurora-m
- ICML WS'26* 6. Gathering Context that Supports Decisions via Entropy Search with Language Models.
 • Closed 85% of performance gap to fully-informed agents via uncertainty-driven information seeking.

1. Xaira Therapeutics – *AI Resident '26–Present*

- De Novo Design Team

2. Stanford University – *Visiting Researcher '26*

- Post-trained AlphaFold 3 via reinforcement learning, SOTA on structure fidelity.
- Achieved optimal drug screening on large libraries via Bayesian optimization.

3. UMass Amherst – *Research Assistant '23*

- Engineered a multi-modal retrieval system for VQA task, boosting retrieval accuracy by 5%.
- Enhanced policy specialization by optimizing discount factors within the OLS Convex Coverage Set.

4. Ontocord – *Research Intern '23*

- Distilled 7B LLM model to **5x** smaller size while maintaining performance parity.
- Developed open-source Vietnamese LLM by processing 1TB of data.

5. EOG Resources – *Software Engineer Intern '23*

- Built a graph-based visualization tool that accelerated complex data analysis workflows by 10%.
- Automated CI/CD pipelines and enhanced security by migrating repositories to GitHub Actions.

6. VietAI – *Research Intern '22*

- Developed SOTA En-Vi model and improved Biomedical NMT via self-training.
- Drove 6% BLEU improvement in Biomedical NMT via self-training and released Vi-MedNLI dataset.

7. FPT Software – *Research Intern '21*

- Developed an interactive AI frontend, streamlining workflows for 50+ engineers.

1. Multi-Objective GFlowNet for Drug Design

- Generated diverse, synthesizable molecules with optimized affinity and ADMET via SynFlowNet.

Paper Awards: ICLR'25 Workshop Spotlight, ICML'25 Workshop Spotlight

Grants: Google-HAI Grant '24 (\$90,000), Google-HAI Grant '25 (\$100,000), Stanford Marlowe Grant.

Frameworks: PyTorch, Flax, JAX, TensorFlow, Hugging Face, Flask, Node.js, Neo4j

Languages: Python, Java, JavaScript, C/C++, SQL

Tools: Git, Linux, GCP, Slurm, Docker, GitHub, Kubernetes